Fiscal Year:	FY 2021	**Task Last Updated:**	FY 05/03/2021
PI Name:	Bellas, Evangelia Ph.D.		
Project Title:	Effect of Microgravity on Adipose Tissue Remodeling and Function		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
PI Email:	evangelia.bellas@temple.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	215-204-3301
Organization Name:	Temple University		
PI Address 1:	1947 N 12th St		
PI Address 2:	College of Engineering, Bioengineering		
PI Web Page:			
City:	Philadelphia	**State:**	PA
Zip Code:	19122-6018	**Congressional District:**	2
Comments:			
Project Type:	Ground,New Investigation	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
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Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
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Key Personnel Changes/Previous PI:	May 2021 report: no changes		
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Task Description:	The goal of this New Space Biology Investigator proposal is to achieve a working model and framework of adipose tissue remodeling and function under microgravity and recovery conditions. The knowledge gained from this 1 year study will be the foundation for larger scope, multi-year grants. While the research emphasis is on Molecular and Cellular Biology, follow-up proposals would exist at the interface of the Animal Biology Element, Developmental, Reproductive and Evolutionary Biology Element, and the Cross-Cutting-Systems Biology Omics and GeneLab Element. This proposal would yield insight on the effects of different microgravity doses and recovery on metabolic function and remodeling of adipose tissue. The effects of space travel and ground based models of space travel have been well documented in several key human physiological systems. Namely, effects on the musculoskeletal system and systems sensitive to extreme changes in hydrostatic pressure, such as brain, heart, and ocular vessels have been well studied. The musculoskeletal system is well studied because of the connection between mechanical loading and tissue function. Chronic exposure to unloading/microgravity results in a loss of mass and strength of both bone and muscle. Similarly, acute and chronic exposure to microgravity leads to a disruption and redistribution of blood flow and alterations to the immune system. These changes are thought to lead to vascular pressure and hypertension among other side effects. Yet, the effects of space travel on an organ system which maintains whole body metabolic homeostasis via the body's vasculature and immune system has not been studied. Adipose tissue systemically regulates whole body homeostasis (and dysregulation) with coordinated release of cytokines (adipokines) and immune signals. On a tissue level, these signals are part of the complex cross-talk between adipocytes and their microenvironment. To address the knowledge gap in adipose tissue function in response to microgravity and recovery, we will pursue two main research objectives, encompassing microenvironmental remodeling and vascularization in engineered adipose tissue. In this study, we will develop a 3D in vitro model of adipose tissue for culture under microgravity conditions to mimic spaceflight. Rotating wall vessels allow us to generate a weightlessness condition and these will be compared to standard (Earth gravity) dynamic cultures. By first evaluating the effect microgravity has on 3D adipocytes alone, we can isolate the metabolic effects independent of vascular flow. Upon characterization on the effect on adipocytes alone, we can begin to add complexity by co-culturing with endothelial cells to provide a vascular network. This allows us to study how vascular remodeling due to changes in flow and/or pressure can impact metabolic function. Upon conclusion of this one year period, this project will generate a fundamental knowledge base and working model of (vascular) adipose tissue function under microgravity conditions. These findings will have major implications on space travel and human health on Earth. This will provide preliminary data for several larger scope grants as these models can be probed in more detail for mechanistic understanding of the effects of gravity on human metabolic function. The studies can also be expanded to determine how nutrients are stored and metabolized in adipose tissue differently in space and on Earth by supplementing the media with various fatty acids (unsaturated versus saturated) and glucose levels (low versus high). Future directions would also include how obese adipose tissue behaves differently as a result of microgravity exposure and how microgravity may inhibit or enhance browning (thermogenic capacity) of white adipose tissue.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These results will provide insight not only to those who undergo space travel but also to those on Earth. Understanding how acute and chronic changes in mechanical forces influence adipose tissue function may yield new approaches to treating diseases such as type II diabetes and obesity.
Task Progress:	This proposal sought to determine the effect of microgravity on engineered adipose tissue on adipose tissue function and remodeling. We hypothesized lack of Earth's gravity forces would lead to a decrease in cytoskeletal tension (via actin reorganization) and would increase adipocyte function. Unlike differentiation in many other cell types, adipogenesis requires a decrease in cytoskeletal tension, to allow for lipid droplet accumulation. As a result of the decreased cytoskeletal tension, we hypothesized this would lead to decreased remodeling in the microenvironment. We have found that in microgravity, adipocyte gene expression trended towards increased adipogenesis. Basal lipolysis rates increased in microgravity at 2 and 3 weeks in culture. After 4 weeks in culture, adipocytes in microgravity are more mature (i.e., larger and contain more lipids). To characterize the effect of microgravity on classic mechanotransduction pathway (RhoA/ROCK), we quantified total RhoA protein levels, Yes-associated protein (YAP) translocation and mechano-sensitive pathway genes (RhoA, ROCK, YAP1), actin (ACTA2) expression. Interestingly, mechanotransduction outcomes trended towards increased levels in microgravity at both 1 week and 4 week timepoints. Extracellular matrix (ECM) remodeling genes, including collagen 1, fibronectin, and MMP2 (which can degrade collagen type IV and fibronectin), trended towards increased levels in microgravity. Collagen I immunostaining did not show any notable changes in deposited collagen I at 1 or 4 weeks. Fibronectin staining is underway. However, integrin beta 1 (links ECM to cell) gene expression was increased in microgravity. Cytoskeletal organization was broken down into 3 bins across two cell “types.” Actin staining was quantified in image analysis for lipid droplet (LD) positive cells (i.e., mature adipocytes) versus cells that do not display strong LD staining. These cell types where then broken into groups that expressed cytoplasmic actin, cortical actin (c-actin), or actin stress fibers (f-actin). The results thus far have partly confirmed our hypothesis, in that adipocyte maturation will increase in microgravity; however, not as a result of decreased cytoskeletal/actin tension. We are exploring possible mechanisms linking the actin reorganization and adipocyte maturation.
Bibliography Type:	Description: (Last Updated:)